Hamza Boulahia

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WORK EXPERIENCE

AI Consultant Freelancer
(May 2024 – Present)

AI Inheritance Chatbot
- Led dataset creation and preprocessing for domain-specific Islamic inheritance Q&A, preparing data for both fine-tuning and RAG.
- Experimented with quantized LoRA fine-tuning of open-source LLMs using augmented Arabic data; evaluated via ROUGE, BLEU and custom LLM-based metrics, and concluded RAG was preferable.
- Benchmarked Arabic embedding models to select the best for vector indexing.
- Designed, implemented, and evaluated multiple RAG pipelines (simple RAG, reranking, LLM-judged), using classical NLP, custom LLM, and RAG-specific metrics to identify the optimal setup.
- Integrated the chosen RAG retrieval module into the chatbot to deliver accurate, context-aware inheritance answers in Arabic.
  - **Tech stack:** Python, LangChain, FAISS, Hugging Face Transformers, PyTorch, unsloth, ragas, Gradio.

Customer Service Chatbot
- Built a customer service chatbot leveraging Retrieval-Augmented Generation (RAG).
- Implemented vector search with Pinecone to retrieve relevant knowledge base content.
- Developed a fallback escalation mechanism to forward complex requests to human agents via email integration.
  - **Tech stack:** Python, LangChain

Data Scientist - Aléria Consulting
(May 2022 – May 2024)
- Developed a web application for webpage attention heatmap prediction
  - Training and testing a deep learning model to predict webpage saliency.
  - Achieving more than 85% accuracy on the Webpage Saliency public dataset.
  - Implemented a robust Python API for seamless model integration.
    - **Tech stack:** Python, U-Net, OpenCV, Flask
- Designed a Windows desktop application for ID Optical Character Recognition (OCR)
  - Creating and optimizing object detection models for efficient extraction of information from ID cards.
  - Applied advanced image processing techniques to minimize OCR errors.
  - Achieving accurate and reliable extraction of data from ID cards, and delivering product within due time.
    - **Tech stack:** Python, YOLO, OpenCV, Tesseract-OCR, PyQT5

Data Science Consultant Freelancer
(Oct 2020 – May 2022)
- Creating and improving classification and regression machine learning models using Python as per client's requirements.
- Performing exploratory data analysis to extract insights and delivering detailed reports.

Math Tutor
Sigma Training Center (Oct 2016 – Aug 2018)
- Tutoring students in bachelor level mathematics (Calculus, Linear Algebra, and Convex Optimization).
- Incorporating visuals using Python to help students better understand certain math concepts.

EDUCATION

Research Master in Mathematical Modeling and Scientific Calculations
Relevant subjects: Statistics, Stochastics, Optimization
National Engineers School of Tunisia (2014 – 2016)

Bachelor of Science in Applied Mathematics
Relevant subjects: Linear Algebra, Calculus, Statistics, Algorithmics & Data Structures
Higher School of Science and Technology of Tunis (2011 – 2014)

CERTIFICATIONS

- TensorFlow 2 for Deep Learning (Coursera - Aug 2021)
- Deep Learning Specialization (Deeplearning.ai - Oct 2020)
- Machine Learning (Coursera – Apr 2020)
- Data Analyst Nanodegree (Udacity | Oct 2019 – Jun 2020)

SKILLS

AI & Machine Learning
LLM training/fine-tuning, AI agents (LangChain, LangGraph, LlamaIndex), prompt engineering, LLM evaluation (LLM-as-a-Judge)
Natural Language Processing (NLP) with BERT and spaCy
Computer Vision with OpenCV, YOLO models, and Segment Anything (SAM)

RAG Systems
RAG Systems End-to-end RAG pipelines (retriever–reranker–generator), hybrid search, query expansion, context compression
RAG evaluation (faithfulness, answer relevance, retrieval precision/recall), and custom metrics

Programming & ML Frameworks
Python (Pandas, Scikit-learn, Xgboost, PyTorch, TensorFlow, FastAPI, Flask, Plotly, Streamlit, Gradio)

Tools & DevOps
Git, CI/CD with GitHub Actions, Docker, AWS, Web automation and scraping with Selenium and Playwright

Databases & Systems
SQLite, MySQL, SQL-based data systems
Desktop and UI applications with PyQt5

Languages
Arabic (Native), French (Professional), English (Professional)

PROJECTS & CONTENT CREATION

- **Medium: Hamza Boulahia**
  - Author of a technical AI blog with 30+ published articles and 1,100+ followers.
  - Write in-depth content on Large Language Models (LLMs), AI agents, Retrieval-Augmented Generation (RAG), and modern AI systems.
  - Publish implementation guides, projects, and technical experiments, including topics such as loss landscapes, embeddings, and transformer architectures.
- **YouTube Channel: Machine Learning With Hamza**
  - Creator & Host of ~32 educational videos focusing on machine learning and AI tutorials; channel grew to ~5.5K subscribers as of June 2025
  - Produced content on topics such as neural networks, fine-tuning LLMs, chatbots, NLP and CV techniques, and Reinforcement Learning.
  - Managed end-to-end production: topic research, scripting, recording, editing, and thumbnail design; solicited and responded to viewer questions, fostering community engagement and feedback loops.

- **Web application for football video analysis** (Github | YOLOv8, OpenCV, Streamlit):
  Creating a web application for football object detection with team prediction and tactical map representation.
- **Customized chatbot web application** (Github | GPT-3.5, Gradio, Tacotron 2):
  Creating a chatbot web application for lyrics generation and speech synthesis using deep learning model for voice cloning.
- **Web scraper for job offers on LinkedIn** (Github | Python, Playwright, Github Actions):
  Creating a web scraper to collect job offers data based on a job title and location, and save the extracted data in csv files.

INTERESTS

Reading, educational videos creation, and calisthenics workout.